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Medical Expert Systems

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Pneumonia Diagnosis System

Pediatric Pneumonia Diagnosis System

Upload Chest X-ray

Age:

Gender:

☐ Male ☒ Female

Select Symptoms (Ctrl/Shift to select multiple):

Fever
Cough
Shortness of Breath
Chest Pain
Fatigue
Sore Throat

Diagnose

Diagnosis:

Clear

Pediatric Pneumonia Diagnosis System

Upload Chest X-ray



Age:

10

Gender:

☐ Male ☒ Female

Select Symptoms (Ctrl/Shift to select multiple):

Fever
Cough
Shortness of Breath
Chest Pain
Fatigue
Sore Throat

Diagnose

Diagnosis: Normal (Severe Pneumonia)
Risk Score: 50

No treatment needed. Maintain general health precautions and monitor symptoms. Severe treatment may be needed, including IV antibiotics or oxygen support.

Clear

➤ **Pneumonia in children**

- A lung infection causing inflammation in the air sacs, leading to symptoms like fever, cough, and shortness of breath. It can be caused by bacteria, viruses, or fungi and is particularly dangerous for vulnerable groups.
- Early detection through chest X-rays is crucial for effective treatment and preventing complications.

➤ **Data Used**

- Capturing high-quality pediatric chest X-rays is challenging due to children's discomfort and different physiology. Most datasets focus on adults, highlighting the need for pediatric-specific X-ray classification models.
- [5,856 Chest X-rays labelled as either pneumonia or normal.](#)

➤ **Function of system**

- **The Pneumonia Diagnosis System** leverages deep learning, specifically Convolutional Neural Network (CNN), and a rule-based expert system to diagnose pneumonia from chest X-ray images and refine the diagnosis using pediatric data.
- **This hybrid system** enhances diagnostic accuracy by combining image analysis with pediatric -specific information, offering a more comprehensive assessment of the pediatric condition.

➤ **How the system works**

Setting Dataset Paths and Image Parameters

```
# Dataset paths
train_dir = 'C:/Users/maram/Desktop/project medical expert/train'
test_dir = 'C:/Users/maram/Desktop/project medical expert/test'

# Image dimensions and parameters
IMG_HEIGHT = 150
IMG_WIDTH = 150
BATCH_SIZE = 32
EPOCHS = 10
✓ 0.0s
```

Parameters are for training a CNN on the dataset. The images are resized to **150x150 pixels**, processed in batches of **32 images**, and the model will train for **10 epochs**.

Data Augmentation and Preprocessing for Training and Testing

```
# Data augmentation and rescaling
train_datagen = ImageDataGenerator(rescale=1./255, shear_range=0.2, zoom_range=0.2, horizontal_flip=True)
test_datagen = ImageDataGenerator(rescale=1./255)
```

Rescales pixel values to the range [0, 1] helps improve convergence during training.

shear range=0.2 improve model generalization by making the model robust to small distortions.

horizontal flip=True helps the model learn features regardless of left-right orientation.

```
train_generator = train_datagen.flow_from_directory(
    train_dir,
    target_size=(IMG_HEIGHT, IMG_WIDTH),
    batch_size=BATCH_SIZE,
    class_mode='binary'
)

test_generator = test_datagen.flow_from_directory(
    test_dir,
    target_size=(IMG_HEIGHT, IMG_WIDTH),
    batch_size=BATCH_SIZE,
    class_mode='binary'
)
```

creates a data generator for training by loading images, resizing them to the given target_size, batching them with the specified batch_size, and labeling them for binary classification (pneumonia vs. normal) using class_mode='binary'.

creates a data generator for the test set by loading images, resizing them, batching them with the specified BATCH_SIZE, and labeling them for binary classification using class_mode='binary'. It uses test_datagen for image preprocessing, which only rescales the images.

Building the Convolutional Neural Network (CNN) Model

```
# Build the CNN model
model = Sequential([
    Conv2D(32, (3, 3), activation='relu', input_shape=(IMG_HEIGHT, IMG_WIDTH, 3)),
    MaxPooling2D(pool_size=(2, 2)),
    Conv2D(64, (3, 3), activation='relu'),
    MaxPooling2D(pool_size=(2, 2)),
    Flatten(),
    Dense(128, activation='relu'),
    Dropout(0.5),
    Dense(1, activation='sigmoid')
])
```

Convolutional layers extract features from the images. The first one has 32 filters and the second has 64 filters.

Fully connected layers with 128 neurons and ReLU activation to learn high-level features.

reduce the spatial dimensions of the feature maps

Converts the 2D output of the convolutional layers into a 1D array.

Regularization technique that randomly disables 50% of neurons to avoid overfitting.

The final layer with a sigmoid activation function outputs a probability for binary classification (0 or 1).

```
# Compile the model
model.compile(optimizer=Adam(learning_rate=0.001), loss='binary_crossentropy', metrics=['accuracy'])

# Train the model and store the history for plotting
history = model.fit(train_generator, epochs=EPOCHS, validation_data=test_generator)

# Save the trained model
model.save('pneumonia_cnn_model.h5')
```

used as the loss function for binary classification

adjust the weights of the model based on the gradients.

tracked a performance metric during training

Model Evaluation and Accuracy Calculation

```
# Evaluate on train and test data
train_loss, train_acc = model.evaluate(train_generator, verbose=0)
test_loss, test_acc = model.evaluate(test_generator, verbose=0)

# Get the number of samples in train and test datasets
train_samples = train_generator.samples
test_samples = test_generator.samples
total_samples = train_samples + test_samples

# Calculate overall accuracy
overall_acc = (train_acc * train_samples + test_acc * test_samples) / total_samples

# Print train, test, and overall accuracy
print(f"Train Accuracy: {train_acc:.2f}")
print(f"Test Accuracy: {test_acc:.2f}")
print(f"Overall Accuracy: {overall_acc:.2f}")
```

Accuracy

Train Accuracy: 0.96
Test Accuracy: 0.86
Overall Accuracy: 0.95

Plotting Training and Validation Loss Curve

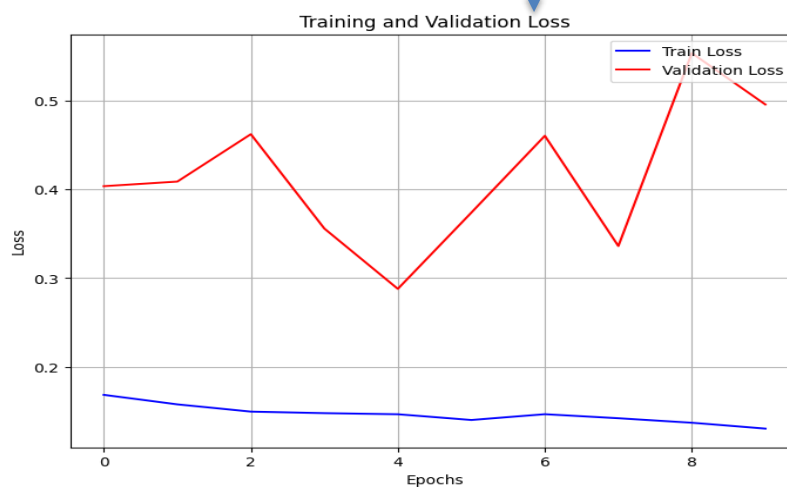
```
# Function to plot the loss curve
def plot_loss_curve(history):
    loss = history.history['loss']
    val_loss = history.history['val_loss']
    epochs_range = range(len(loss))

    plt.figure(figsize=(8, 6))

    # Plot loss
    plt.plot(epochs_range, loss, label='Train Loss', color='blue')
    plt.plot(epochs_range, val_loss, label='Validation Loss', color='red')
    plt.xlabel('Epochs')
    plt.ylabel('Loss')
    plt.title('Training and Validation Loss')
    plt.legend(loc='upper right')

    plt.grid(True)
    plt.show()

# Call the function to plot the loss curve
plot_loss_curve(history)
```



Implementing Rule-Based Diagnosis System

```
def rule_based_diagnosis(age, gender, symptoms, cnn_prediction):  
    if cnn_prediction == 1:  
        diagnosis = "Pneumonia Detected"  
        treatment_plan = "Immediate medical attention required. Antibiotics may be prescribed. Consider hospitalization based on severity."  
    else:  
        diagnosis = "Normal"  
        treatment_plan = "No treatment needed. Maintain general health precautions and monitor symptoms."  
  
    # Severity based on symptoms  
    risk_score = len(symptoms) * 10  
    if len(symptoms) <= 2:  
        diagnosis += " (Early Stage)"  
        treatment_plan += " Early intervention recommended."  
    elif len(symptoms) >= 3:  
        diagnosis += " (Severe Pneumonia)"  
        treatment_plan += " Severe treatment may be needed, including IV antibiotics or oxygen support."  
  
    # Additional rule for young children (age < 5)  
    if age < 5 and cnn_prediction == 1:  
        diagnosis = "Severe Pneumonia Risk (Young Age)"  
        treatment_plan = "Immediate medical attention required. Oxygen therapy and close monitoring needed."  
        risk_score += 20  
  
    return diagnosis, treatment_plan, risk_score
```

Function adjusts the diagnosis and treatment plan based on the CNN prediction and additional factors like symptoms and age

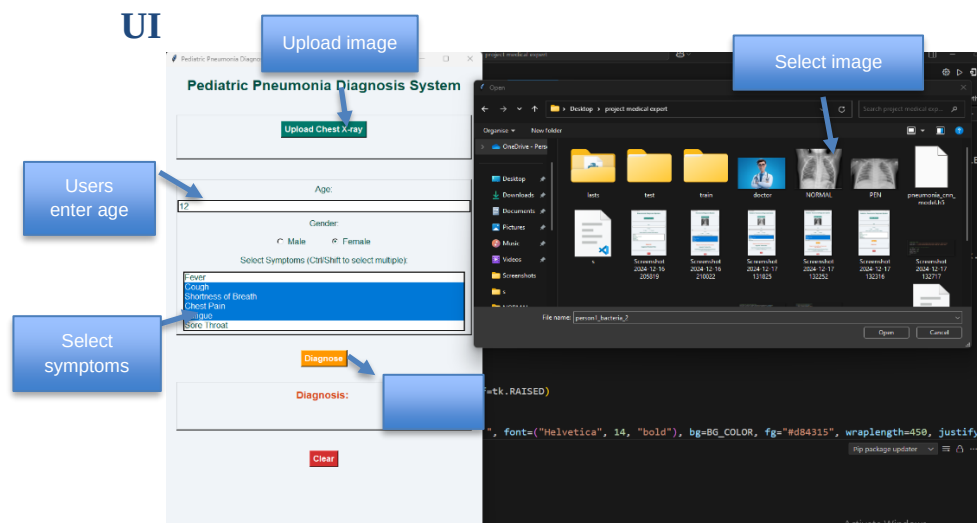
CNN Prediction = 1 (Pneumonia):

- Diagnosis: "Pneumonia Detected"
- Treatment Plan: Immediate medical attention; antibiotic
- Severity: If symptoms are few (≤ 2), it's "Early Stage"; if many (≥ 3), it's "Severe"
- Age < 5: Higher risk, with emphasis on immediate care and increased risk score.

CNN Prediction = 0 (Normal):

- Diagnosis: "Normal"
- Treatment Plan: General health monitoring; no treatment required.
- Severity: Based on the number of symptoms

This system combines CNN predictions with rules based on symptoms and age to refine the diagnosis and recommend appropriate treatments.



Output

Pediatric Pneumonia Diagnosis System

Upload Chest X-ray



Age: 12

Gender: ☐ Male ☒ Female

Select Symptoms (Ctrl/Shift to select multiple):

- ☐ Fever
- ☐ Cough
- ☐ Shortness of Breath
- ☐ Chest Pain
- ☐ Fatigue
- ☐ Sore Throat

Diagnose

Diagnosis: Normal (Severe Pneumonia)
Risk Score: 40

No treatment needed. Maintain general health precautions and monitor symptoms. Severe treatment may be needed, including IV antibiotics or oxygen support.

Clear

Normal case has early detection

Pediatric Pneumonia Diagnosis System

Upload Chest X-ray



Age: 9

Gender: ☐ Male ☒ Female

Select Symptoms (Ctrl/Shift to select multiple):

- ☒ Shortness of Breath
- ☒ Chest Pain
- ☒ Fatigue
- ☒ Sore Throat
- ☒ Muscle Pain
- ☒ Chills

Diagnose

Diagnosis: Pneumonia Detected (Severe Pneumonia)
Risk Score: 70

Immediate medical attention required. Antibiotics may be prescribed. Consider hospitalization based on severity. Severe treatment may be needed, including IV antibiotics or oxygen support.

Pneumonia case

